

[OP] AI Self-Evolution Direction and Neo lab Review (Continuously Updated)

Neo lab: A new type of AI-native startup initiated by top AI researchers/engineering teams, characterized by a small team, high computing power, and strong research intensity.

1. How to understand AI self-evolution/recursive self-improvement?

- **AI for AI:** Using AI to automate the R&D, training, evaluation, deployment, monitoring, and iterative processes of AI systems, which in essence is using AI to transform the production function of AI. It does not necessarily form a closed-loop of capabilities.
- **AI Self-Evolution:** AI systems utilize themselves or other AI systems to identify defects, propose improvements, conduct experiments, evaluate results, and feed effective improvements back into the next version of the system.
 - **Core question: Can AI participate in or even lead the process of continuously improving its own capabilities?** More specifically, can AI continuously update its models, memories, tools, workflows, or architectures based on its own experience, task trajectories, or environmental feedback to enhance future performance .
- **The essence of self-evolution: Model (intelligence), Agent (action ability), Environment (feedback and verification) The closed loop of co-evolution**
 - Closed-loop maturity assessment: L3 is the currently verifiable inflection point, L4 is the key to forming a flywheel, and L5 remains highly uncertain (the long-term breakthrough direction of leading model companies).

Level	Morphology	Definition	Judgment Criteria
L0	Single-point AI Tool	Only completes tasks without self-improvement	Is it just a one-time output?
L1	Self-correction within a single task	Optimize a specific process, iterate within a single task/workflow, generate → critique → revise	Whether to improve the performance of the current task For example, the coding agent automatically fixes bugs

L2	Cross-task experience accumulation	memory / tools / workflow can be reused	Whether to change future task performance
L3	Automated Research Closed Loop	hypothesis → experiment → feedback → next hypothesis/future research direction	Whether one can conduct experiments independently
L4	Capacity Recirculation Closed Loop	Experience enters the next generation model or agent	Whether a data/model flywheel has been formed
L5	Open-ended recursive self-improvement	The system continuously enhances its self-improvement capabilities	Whether it can continuously discover new improvement directions and generalize across tasks and environments

2. Why focus on AI self-evolution?

- **Vision: One of the core paths leading to superintelligence.**
 - AI self-evolution is the A-side of superintelligence, exploring whether a single or a group of AI systems can continuously improve their own capabilities and break through the boundaries of human cognition
 - Swarm intelligence and multi-agent collaboration are the flip side, exploring how multiple AI systems can generate higher-order intelligence through collaboration, competition, and division of labor
- **Reality: The marginal revenue of model pre-training is gradually diminishing, and it is necessary to further explore the path to enhance the boundaries of model intelligence; one of the directions of the new scaling law (from scaling the number of parameters × data volume × computing power to scaling the learning process)**

Since the breakthrough in Deep learning, the improvement of AI capabilities has mainly relied on:

Stage	dominant paradigm	Core Assets
2012–2020	Industrialization of Deep Learning	Data, GPU, Model Architecture

2020–2024	Platformization of Foundation Model	Pre-trained models, RLHF, application ecosystem
2024–2027	Agentic AI Engineering	Agent, Tool Invocation, Workflow, Code Generation
2026+	AI Recursive Self-Evolution	AI participates in AI R&D, forming a self-evolving closed loop

The current industry is facing issues such as the scarcity of high-quality human data, rising pre-training costs, and the decline of marginal revenue as the parameter scale continues to expand. AI self-evolution offers a new direction for scaling law: shifting from scaling the model to scaling experience, feedback, experiments, discovery, and the self-improvement loop.

- Application Prospects: Complex Science and Engineering Problems**

AI self-evolution is suitable for fields such as drug R&D, new materials, new energy (nuclear fusion), chip design, robotics, etc., that require **complex search, complex simulation, complex experimental verification, and multi-objective optimization**. These fields have a vast search space, a scarcity of human experts, and high trial-and-error costs, but once an automated research closed-loop is formed, both the commercial and scientific value are extremely high.

Field	Why is it suitable for AI self-evolution
Drug R&D	The molecular space is vast, with slow experimental feedback but high value
New Materials	The combinatorial space is extremely large, requiring a closed-loop of prediction, simulation, and experiment
New Energy / Battery	Multi-objective optimization, where performance, cost, safety, and lifespan are mutually constrained
Nuclear Fusion	Complex physical systems require control, simulation, and real-time optimization
Chip Design	The design space is vast, and the multi-objective constraints of PPA are strong
Robot	Needs to learn from trial and error and environmental interaction
Math / Code	Digitally verifiable, suitable for rapid self-improvement

3. Self-evolving exploration path: experience closed-loop + systematic evolution

3.1 Experience Closed Loop: How Does AI Continuously Strengthen Itself from Feedback?

Overview of LLM Self-Evolution summarizes self-evolution as a four-stage cycle: **experience acquisition, experience refinement, updating, and evaluation**; among them, experience acquisition includes the acquisition of tasks, solutions, and feedback, and updating includes two types: in-weight and in-context updates.

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1

Experience Acquisition

2

→ Experience Refinement

3

→ Updating

4

→ Evaluation

5

→ Next Iteration

Phase	Meaning	Industrialization Mapping
Experience Acquisition	Obtain tasks, solutions, trajectories, failure samples, and feedback	synthetic data, agent trajectory, research trace, experimental data platform
Experience Refinement	Filter out noise and determine which experiences are worth learning	eval infra、AI judge、reward model、verifier
Updating	Absorb experience, update models, memories, tools, or workflows	post-training、RL infra、memory update、workflow optimizer
Evaluation	Determine whether the system has truly become stronger	benchmark, code testing, simulation, real experiment, industrial indicator

3.2 Systematic Evolution: Which parts of an AI system can evolve?

Refer to [self-evolving agents](#) review, the objects of evolution include not only model weights but also other components of the agent system.

Dimension	Core Issue	Content
What to evolve	What is evolution?	Model、Memory / Context、Tools、Architecture
When to evolve	When will it evolve?	Intra-test-time, Inter-test-time (accumulating experience across tasks and changing future behavior), Long-horizon continual evolution
How to evolve	How to evolve?	Feedback sources: Reward, Verifier Improvement methods: Reward-based (RL), Imitation/demonstration, Population-based methods (e.g., Evolutionary Search)
Where to evolve	Where does it evolve?	General intelligent agents, as well as vertical scenarios such as code, GUI, finance, healthcare, education, etc.

3.3 Three types of paths (classified by feedback closed-loop, not mutually exclusive, and are three capability modules of the self-evolving system)

路径	解决的问题	典型方向	本质
模型自改进	模型怎么变强？	自反馈、自训练、AI 监督、verifier、test-time reasoning、post-training	改进模型训练与推理，模型或推理
Agent / Research 自改进	AI 系统怎么更好地行动和研究？	终身学习、多智能体、AI Scientist、research agent、coding agent	改进任务执行与研究流程，AI 系统 researcher，持续积累经验、提出 workflow。
环境驱动自改进	AI 如何通过环境获得硬反馈？	reward / environment design、机器人仿真、自主实验室、AI4S	通过环境、仿真或真实实验获得硬识。

- **Evolutionary Path :**
 - **Horizontal:** First, develop AI into a "Machine Researcher / AI Scientist", enabling AI to autonomously complete the research closed-loop from hypothesis formulation, experimental design, experiment execution, result analysis to conclusion formation in an evaluable environment; then transfer this closed-loop capability to complex scientific and engineering problems such as drugs, materials, energy, chips, and robots.

- **Vertical:** AI Scientist generates experimental trajectories and feedback → absorbs these experiences during/after training → promotes the improvement of model capabilities → stronger models support stronger AI Scientists → ultimately may approach a model paradigm revolution

4. Open question

The market ceiling for AI self-evolution does not only come from efficiency tools (substituting/enhancing existing R&D expenditures), but from the AI R&D production function, scientific R&D processes, and new asset creation.

Challenge	Key Issues
Feedback Quality	LLM self-assessment and AI judge are unstable; truly reliable feedback comes from code, mathematics, simulation, real experiments, or industrial results (hard feedback is scarce); from hard feedback to open exploration
Evaluation Dynamization	需要 long-horizon、 adaptive benchmark
Stability-Plasticity Dilemma	The system needs to absorb new experiences without catastrophic forgetting, memory pollution, or capability drift
Reward hacking / mis-evolution	The system may continuously optimize the wrong objectives and solidify the deviations into memory, tools, or workflows
Sustainable Improvement Space	One-time optimization does not constitute self-evolution; the key lies in whether it can achieve continuous multi-round improvement
Security and Governance	When AI can modify memory, tools, workflow, or call external environments, permission control, sandbox, rollback, and auditing are required

- **From hard feedback to open exploration**

AI self-evolution relies on feedback, but **feedback is not always immediate, objective, or verifiable**. In the short term, the most easily achievable are strong feedback environments such as code, mathematics, simulation, and laboratory settings; however, **the key tasks leading to superintelligence are often open-ended problems**, such as new scientific theories, long-term

research directions, complex strategic judgments, and the generation of original concepts. Feedback for such tasks is weak, delayed, and sometimes even not fully verifiable.

The strong self-evolution system not only needs to utilize verifiable feedback but also must possess **the ability to break down unverifiable problems into verifiable subproblems** , including generating falsifiable predictions, designing experiments, constructing proxy metrics, introducing multi-agent refutation, recording uncertainties, and retaining audit trails.

Level	Feedback Type	Example	Risk
V5 Strong Verification	Objective, Immediate, Repeatable	Code testing, mathematical proof checker	The task space may be narrow
V4 Experimental Verification	Real World Experiment	Drugs, Materials, Robots	Slow, expensive, and heavy asset
V3 Expert Verification	Human Expert / Peer Review	Paper review, medical opinion	Subjective, slow, and biased
V2 Proxy Indicator	citation, user adoption, benchmark, market signal	Business adoption, paper influence	Goodhart's Risk
V1 Internal Consistency	Multi-model mutual evaluation, logical self-consistency, debate	AI judge、debate	Self-reinforcing error
V0 cannot be verified / Forward verification	New theory, long-term strategy, open innovation	Original scientific paradigm	Hallucinatory progress poses the highest risk

5. Mapping of Startup Teams

Disclosed domestic and overseas startups with relatively high attention in the current field (classified by the maturity level of self-evolving closed-loop)

Category	represents the company	Core Closed Loop	Close d-loop Maturity	Key Verification	Market Opportunities and Risks

Recursive / Superlearner	Recursive, Ineffable	AI enhances its own capabilities through self-improvement or learning from experience	L4–L5	Whether it can generate cross-task, long-term ability growth	Odds are the highest, but verification is the most difficult and valuation is the easiest to frontload - Base TAM (Current Payable Budget): Not clear in the early stage - Upside TAM: AI R&D engine / foundation AI platform - Option TAM (Paradigm Breakthrough or Asset Creation Value): New Intelligence Paradigm
AI-for-AI Lab Automation	Core Automation, Autoscience, Analemma, Sakana	AI Automation AI/ML Research and Experimentation	L3–L4	Whether to improve the real research throughput and allow the results to be fed back	Quick feedback, clear client, close to AI R&D budget - Base TAM: AI/ML Engineering and Research Tools Budget - Upside TAM: AI R&D OS - Option TAM: 自动化 AI lab / model factory
AI Scientist / Autonomous Lab	Periodic, Lila, Isomorphic, Depth Principle	General/vertical domain: AI proposes scientific hypotheses → simulation/experimental verification → proprietary data feedback → next round of discovery	L3–L4	Whether a proprietary experimental data flywheel and monetizable discoveries have been formed	The ceiling is extremely high, but the cycle and capital intensity are greater - Base TAM: AI4S Software and Scientific Research Platform Budget - Upside TAM: Industry R&D Platform - Option TAM: Drug, Material, Energy, and Chip Asset Factory
Reasoning Self-Improvement	Poetiq	Does not change the weights, self-optimizes through the reasoning harness	L1–L2	Whether it generalizes across tasks	Fast productization, LLM inference budget, but barriers and long-term independence need to be verified

					• Base TAM: Model Performance Enhancement Tool • Upside TAM: test-time intelligence orchestration layer • Option TAM: Become the default control layer for model invocation
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Details

Category	Core Closed Loop	Close d-loop maturity	represents the company	Expand description
Recursive / Superlearner	AI enhances its capabilities through self-improvement or learning from experience	L4–L5	Recursive super intelligence	Positioning: Recursive is the purest recursive self-improvement / superintelligence lab, aiming to enable AI to learn to improve itself through open-ended algorithms and drive automated scientific discovery. GV revealed in May that its early funding was \$650M , with a valuation of approximately \$4.65B . Team Background: The company is led by Richard Socher. Co-founders include Tim Rocktäschel. The team, with about 20 members, is from institutions such as OpenAI, Google DeepMind, Meta, etc.; Socher once served as the chief scientist at Salesforce, and Rocktäschel once served as the principal scientist at DeepMind. Research Direction / Product:Currently more like a research-first neo lab, with the core direction being: • self-improving AI • open-ended algorithms • automated knowledge discovery • AI automatically formulates, tests, and validates scientific hypotheses
			Ineffable	

				Positioning: Ineffable is an experience-based superlearner lab founded by David Silver, betting that AI will no longer primarily rely on human-generated data but acquire capabilities through RL, trial and error, and environmental interaction. TechCrunch reported in April that it completed \$1.1B in financing, with a valuation of \$5.1B. Team Background: The core founder is David Silver, a key figure in AlphaGo / AlphaZero and former head of reinforcement learning at Google DeepMind. Wired reported that he recruited top researchers from Google DeepMind and other frontier labs. Research Direction / Product:Public narratives focus on • reinforcement learning (collaborating with NV to build RL-oriented infrastructure) • learning from experience • trial-and-error intelligence • superlearner that does not rely on large-scale human text data
AI-for-AI Lab Automation	AI Automation AI/ML Research and Experimentation	L3-L4	Core Automation	Positioning : Build “the most automated AI lab in the world,” starting with automating its own research. The company was founded by former OpenAI researcher Jerry Tworek, with the goal not to simply continue scaling large models, but to explore new learning algorithms and automate the AI lab. Team Background : Core Automation has attracted researchers from institutions such as Anthropic, Google DeepMind, OpenAI, including Rohan Anil, Anmol Gulati, Joanne Jang, Ehsan Amid, etc. Research Direction / Product: • AI lab automation • AI research workflow automation • New Learning Algorithm • Self-improvement System • Automated experimental design and research execution
			Sakana	定位: Sakana AI 是东京 AI research company, 核心方向是 nature-inspired intelligence、collective intelligence、evolutionary model merging、self-

			evolution、 Japan sovereign AI。 The company's official website in Series A states that it focuses on new foundation models based on nature-inspired intelligence . <i>In 2025</i> , it completed \$135 million Series B <i>financing</i> , <i>with its latest</i> valuation reaching \$2.635 billion . Team Background: Sakana was founded by former Google researcher David Ha and Llion Jones ; Reuters reported that Jones is one of the authors of the Transformer architecture paper, and Ha served as the head of research at Stability AI / a researcher at Google Brain. Research Direction / Product: • evolutionary model merging • collective intelligence • self-evolving models • AI Scientist • Japanese sovereign AI • finance / defense / manufacturing applications According to the official Series B announcement of Sakana, the funds will be used to advance research on collective intelligence and self-evolution. Compared to pure neo labs, Sakana is more pragmatic: it has a sovereign AI narrative and also has implementation directions in Japanese industries such as finance, defense, and manufacturing.
	Autoscience		Positioning:Autoscience is a more clearly productized automated AI research lab. The company claims to automate the R&D, validation, and deployment of new ML models, and its system is used to train SOTA models for frontier and applied AI labs.Its seed round financing of \$ 14M was led by General Catalyst, with participation from Toyota Ventures, Perplexity Fund, MaC Ventures, and S32. Team Background: The company emphasizes that its system consists of non-human AI Scientists and Engineers, used to train specialized SOTA models. Research Direction / Product: • virtual AI laboratory • AI Scientists

				- AI Engineers - Automatically invent, validate, and deploy specialized ML models - Model R&D Automation for Frontier Labs and Applied AI Labs
			Analemma	Positioning : Di Xing Ji is currently China's most typical AI-for-AI research automation company. Its core system is FARS: Fully Automated Research System, with the goal of automatically completing literature research, formulating hypotheses, designing experiments, writing and executing code, analyzing data, and writing papers. It has LIVE deployed FARS and aims to produce 100 papers. It has received tens of millions of dollars in Angel Round Financing from institutions such as Sequoia China, Gao Rong Capital, Light Source Capital, Jiacheng Capital, and Meituan Longzhu. Team Background : Founder Sun Tianxiang , former core developer of the MOSS large model; his personal homepage shows that he is the founder & CEO of Analemma, obtained a computer doctorate from Fudan University in 2024, and has interned at Shanghai AI Lab, Alibaba DAMO Academy, and Amazon Shanghai AI Lab. Research Direction / Product Direction: - AI Automatic Scientific Research - AI research workflow automation - AI4AI - FARS End-to-End Scientific Research System - Automatic paper generation and experiment execution
			Disarray	
AI Scientist / Autonomous Lab	AI proposes scientific hypotheses → simulation/experimental verification → proprietary data backflow	L3–L4	Periodic Labs	Positioning: Building AI scientists and autonomous laboratories. The official website emphasizes that new knowledge needs to be consistent with reality, so they aim to build AI scientists and autonomous labs for them to operate. Completed approximately \$300M seed; Forbes also reported that it was previously in talks for financing, with a valuation of approximately \$7.5B

	→ next round of discovery		Team Background : Public reports indicate that it was founded by former OpenAI and DeepMind researchers, attracting a large number of researchers from institutions such as Meta, OpenAI, and Google DeepMind. Research Direction / Product: • AI scientist • autonomous lab • physical sciences • physics / chemistry / materials discovery • hypothesis → experiment → feedback loop
		Lila Sciences	Positioning: scientific superintelligence / AI Science Factories. The company said it will expand AI Science Factories and deploy the scientific superintelligence platform to partners. Series A has completed \$350M, with cumulative financing of \$550M. Reuters reported that after the new round of extension, the valuation exceeded \$1.3B, with investors including Nvidia's venture arm and others. Team Background: Lila was incubated by Flagship Pioneering, with Geoffrey von Maltzahn as its CEO. Reuters reported that it was founded in 2023, with its core being the combination of specialized AI models and automated laboratories to generate proprietary scientific data through original experiments. Research Direction / Product: • AI Science Factories • robotic labs • proprietary scientific data • scientific superintelligence • enterprise scenarios such as energy, semiconductors, drug development, etc. Reuters reported that Lila plans to monetize by providing enterprise software access to companies in sectors such as energy, semiconductors, and drug development, and has signed a lease for 235,500 square feet of space in Cambridge to expand its experimental facilities.

		Isomorph ic Labs 定位：Isomorphic Labs 是 Google DeepMind spin-off，定位为 AI-first drug design and development company，核心是 AI drug design engine。The company announced in 2025 a \$600M external investment round to develop the next-generation AI drug design engine and advance therapeutic programs into the clinic. In 2026, another \$2.1B funding was reported, to scale the AI drug design engine, global operations, and drug candidate pipeline. Team Background: The company is led by Demis Hassabis and was spun off from DeepMind in 2021. Its technological foundation is closely related to AlphaFold / AlphaFold 3. Fierce Biotech reported that it uses AI models such as AlphaFold 3 to predict molecular structures and interactions and advance the drug design engine IsoDDE. Research Direction / Product: • AI drug design engine • IsoDDE • oncology / immunology • therapeutic pipeline • pharma partnerships Public information shows that it has collaborations with large pharmaceutical companies such as Eli Lilly and Novartis; Wired reported that Isomorphic is close to launching human trials of AI-designed drugs and mentioned the progress of IsoDDE and its drug design platform.
		Depth Principle Positioning : Depth Principle is a representative company of China's AI for Materials direction, and its official website claims that it is committed to accelerating material innovation through AI. In 2025, Deep Principle completed a Pre-A round of over 100 million yuan, with investors including Lenovo Capital and Incubator Group, Lenovo Star, Jinqiu Fund, BV Baidu Ventures, Qigao Capital, etc. By the end of 2025, it completed a Series A round of over 100 million yuan, co-led by Alibaba Entrepreneurs Fund Greater Bay Area Fund and Ant Group. The Series A funds will be used to upgrade Agent Mira,

				advance AI Materials Factory, and promote self-developed pipelines. In 2026, it announced the completion of a Series A2 round, led by Jinma Investment, with additional investments from Jingtai Technology, Qigao Capital, BV Baidu Ventures, etc. Series A2 funds will be used to iterate the "LLM + Diffusion" dual-driven algorithm system and advance Agent Mira and AI Materials Factory. Team Background : The company was incubated by Jingtai Technology / has strong associated background, and focuses on materials chemistry innovation and AI4S. Research Direction / Product: • Agentic AI for Materials Discovery • Agent Mira • AI Materials Factory • LLM + Diffusion Dual-Driven Model System • L4 High-throughput Autonomous Laboratory • Material R&D Pipeline Market progress is mainly reflected in cooperation with leading domestic and international clients, as well as the advancement of material discovery agents and the construction of autonomous laboratories
Reasoning Self-Improvement	Does not change the weights, self-optimizes through the reasoning harness	L1-L2	Poetiq	Positioning: Reasoning Self-Improvement / meta-system Company. The official website states that its goal is "safe super intelligence, paved with better reasoning," and emphasizes that it does not rely on large-scale retraining or traditional RL post-training (because RL is slow, expensive, and requires a large amount of data), but instead builds a self-improving system on top of existing LLMs (directly optimizing the reasoning system on top of existing models). Completed \$45.8M seed funding, founded by Shumeet Baluja and Ian Fischer in 2025. Team Background: Founded by Shumeet Baluja and Ian Fischer. Public information from Y Combinator states that Poetiq was founded by former DeepMind researchers and has achieved significant improvements on the ARC-AGI benchmark by overlaying a recursive self-improvement system on existing models.

				Research Direction / Product: • reasoning harness • meta-system on top of LLMs • recursive self-improvement without fine-tuning • test-time optimization • 自动 prompt / system design / harness search
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What type of team are you Betting on? : Top research taste + able to build high-quality feedback environment + have real workflow entrance + able to precipitate proprietary trajectory/experimental data + improved results can flow back to the next round of system [Need to further abstract to research field/practical experience]

6. Progress of Self-Evolution Exploration by Frontier Model Company

Company / Model / Project	represents trends	corresponds to the self-evolution module	Closed-loop maturity	Investment Research and Judgment
OpenAI / General Inference Model	OpenAI revealed that its model proposed a new construction for the Erdős planar unit distance problem, overturning a core conjecture in discrete geometry; OpenAI stated that the model was not specifically trained for this problem, nor was it a dedicated scaffolded search system.(OpenAI)	Open-ended reasoning; Research / Execution Orchestrator; Mathematical verification	L2-L3	It demonstrates that the general reasoning model has begun to possess the ability of original explorationin cutting-edge mathematical problems. However, this is more like a "single significant discovery," and the continuous self-improvement flywheel has not yet been proven.
Anthropic / Claude	Dario Amodei proposed the concept of "country of geniuses in a datacenter": a large number of high-capability AI instances work in parallel in the Data center, performing tasks that last for	Compute-scale agent swarm; AI-for-AI; Research Orchestrator; Update / Learning Layer	L3-L4 Potential Path	Anthropic represents the path of "scaling AI self-evolution into a Data center-level research organization," rather than a single-point demo. The rumor of "2 million chip

	hours, days, or even weeks. Its public narrative also points to AI's involvement in the R&D, evaluation, training, and security governance of the next generation of AI. (Dario Amodei)			clusters for model self-training."
Google DeepMind / AlphaEvolve	AlphaEvolve is a Gemini-powered evolutionary coding agent that generates candidate algorithms via LLM, then validates, filters, and iterates them through an automatic evaluator, used for mathematics, algorithm discovery, and computational system optimization. (Google DeepMind)	Environment / Feedback Layer; Verifier; Research Orchestrator; Algorithm Evolution	L3	One of the most standard "generate → automatic evaluation → selection → iteration" closed-loop cases to date, demonstrating that the closed-loop of automatic research/automatic engineering in a strong feedback environment is already feasible.
Google / Gemini Co-Scientist	Google Research / DeepMind has launched a multi-agent AI co-scientist based on Gemini, which is used to generate, debate, and evolve scientific hypotheses, and to help formulate research plans; the paper focuses on validating its ability to generate scientific hypotheses in the fields of life science and biomedicine. Google Research	AI Scientist; multi-agent research loop; Hypothesis generation; Scientific feedback	L3	Google is agentizing the scientific discovery process: from literature, hypothesis, refutation, plan to computational discovery. The focus is on organizing the scientific research process into an iterative closed loop.
Sakana AI / The AI Scientist	AI Scientist-v2 generates human peer-reviewed papers through ICLR 2025 ICBINB workshop blind moderation. (Sakana AI), related research has been published in <i>Nature</i> (<i>Sakana AI collaborates with UBC, Vector Institute, Oxford</i>) The system can autonomously generate ideas, search for and read literature, design/program/conduct	AI Scientist; Research / Execution Orchestrator; Automated Reviewer; Verifier; scientific feedback loop	L3, exploring toward s L4	This is one of the most complete public cases of the "autonomous scientific research closed loop" to date. Its value lies not only in writing papers but in breaking down the scientific research process into an agentic loop that is executable, evaluable, and iterable. However, it is currently still mainly limited to computational

	experiments, and write a complete paper after a given research direction;			experiments and suffers from issues such as shallow ideas, insufficient methodological rigor, difficulty in implementing complex code, and errors in citations/charts. (Sakana AI)
Sakana AI / Darwin Gödel Machine	DGM is a self-improving coding agent that can modify its own code and retain effective improvements through programming task evaluations such as SWE-bench and Polyglot; Sakana claims it can continuously improve itself with more compute. (Sakana AI)	Agent architecture self-modification; Verifier; Update Layer; open-ended exploration	L3–L4 Exploration	One of the public research directions approaching "system self-modification". It demonstrates that agent scaffolds / tools / workflows can be systematically searched and improved, but it remains primarily self-modification of coding agents on top of frozen foundation models, not yet recursive self-improvement at the model weight level.
Qwen3.7-Max	Qwen officially disclosed that it completed 1,158 tool calls and 432 kernel evaluations during approximately 35 hours of continuous autonomous execution; third-party reports claim that it achieved approximately 10x geometric mean acceleration. (Qwen Studio)	Code/kernel feedback environment; Tool use; Execution Orchestrator; Engineering optimization closed loop	L3	Typical case of strong feedback self-evolution: clear objectives, objective feedback, automated evaluation, and continuous iteration. It is closer to an "automated engineering researcher".
Meta / Self-Taught Evaluator	Meta proposed the Self-Taught Evaluator, which uses synthetic training data to improve the evaluator and reduce reliance on manual preference annotation. (arXiv)	Refinement / Verifier Layer; AI judge; reward model; self-evaluation	L2–L3	It is not a complete self-evolving system, but it is crucial: if the evaluator can self-improve, the feedback costs of subsequent RL, agent loop, and AI-for-AI will all decrease. It corresponds to the "judgment of what

				experience is worth learning" in the self-evolving closed loop.
MiniMax M2.7	MiniMax officially states that M2.7's internal harness can automatically collect feedback, build internal evaluation sets, and continuously iterate on architecture, skills/MCP implementation, and memory mechanisms. (MiniMax)	Harness self-evolution; Evaluation; Update / Learning Layer; workflow / memory / architecture optimization	L3-L4 Exploration	MiniMax 2.7 is suitable as a case study for "system self-evolution": it does not simply complete tasks, but optimizes its own harness / workflow / memory / eval. However, mainly based on official disclosures and internal evaluations, external verifiability needs to be carefully marked.
ByteDance / Agent-World	Proposed the self-evolving training arena: Automatically explore real-world environment topics, databases, and executable tool ecosystems through Agentic Environment-Task Discovery, synthesize verifiable and difficulty-controllable tasks; then, through Continuous Self-Evolving Agent Training, combine multi-environment reinforcement learning and dynamic task generation to identify capability gaps and conduct targeted training. (arXiv)	Environment / Feedback Layer; Experience Engine; Update / Learning Layer	L3-L4 Exploration	It is a representative of environment-driven self-evolution. It not only trains agents on fixed benchmarks but also allows the environment, tasks, and agent policies to co-evolve, indicating that the training paradigm is shifting from static data to scalable environment + continual self-evolving training.

Reference Materials

A Survey on Self-Evolution of Large Language Models: <https://arxiv.org/html/2404.14387v2#S4>

A Survey of Self-Evolving Agents: What, When, How, and Where to Evolve on the Path to Artificial Super Intelligence: <https://arxiv.org/html/2507.21046v4>

Self-Improvement in Multimodal Large Language Models: A Survey:

<https://arxiv.org/html/2510.02665v1>

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